$$P(\Delta \cdot c \leq 0.767) +$$

$$\frac{S_x^2}{S_y^2} = \Delta \underbrace{\frac{a^2}{b^2}}_{c} = \Delta \cdot c$$

$$+ P(\Delta \cdot c \geq 1.27) = P\left(\Delta \leq \frac{0.767}{c}\right) + P\left(\Delta \geq \frac{1.27}{c}\right)$$

№ 14

$\xi \sim N(a, 2)$ $\quad x = \{-1.11, -6.10, 2.42\}$

$\eta \sim N(b, 1)$ незав. $\quad y = \{-2.29, -2.91\}$

$H_0: a = b$

$H_1: a > b$

(T) Фишера:

$$\frac{\bar{x} - a}{\sqrt{2}} \sqrt{3} \sim N(0, 1)$$

$$\frac{\bar{y} - b}{1} \sqrt{2} \sim N(0, 1)$$

$$\bar{X} - a \sim N\left(0, \frac{2}{3}\right)$$

$$\bar{Y} - b \sim N\left(0, \frac{1}{2}\right)$$

$$\bar{X} - a - (\bar{Y} - b) \sim N\left(0, \underbrace{\frac{2}{3} + \frac{1}{2}}_{7/6}\right)$$

$$\Delta = \frac{\bar{X} - \bar{Y} - (a - b)}{\sqrt{\frac{7}{6}}} \sim N(0, 1)$$

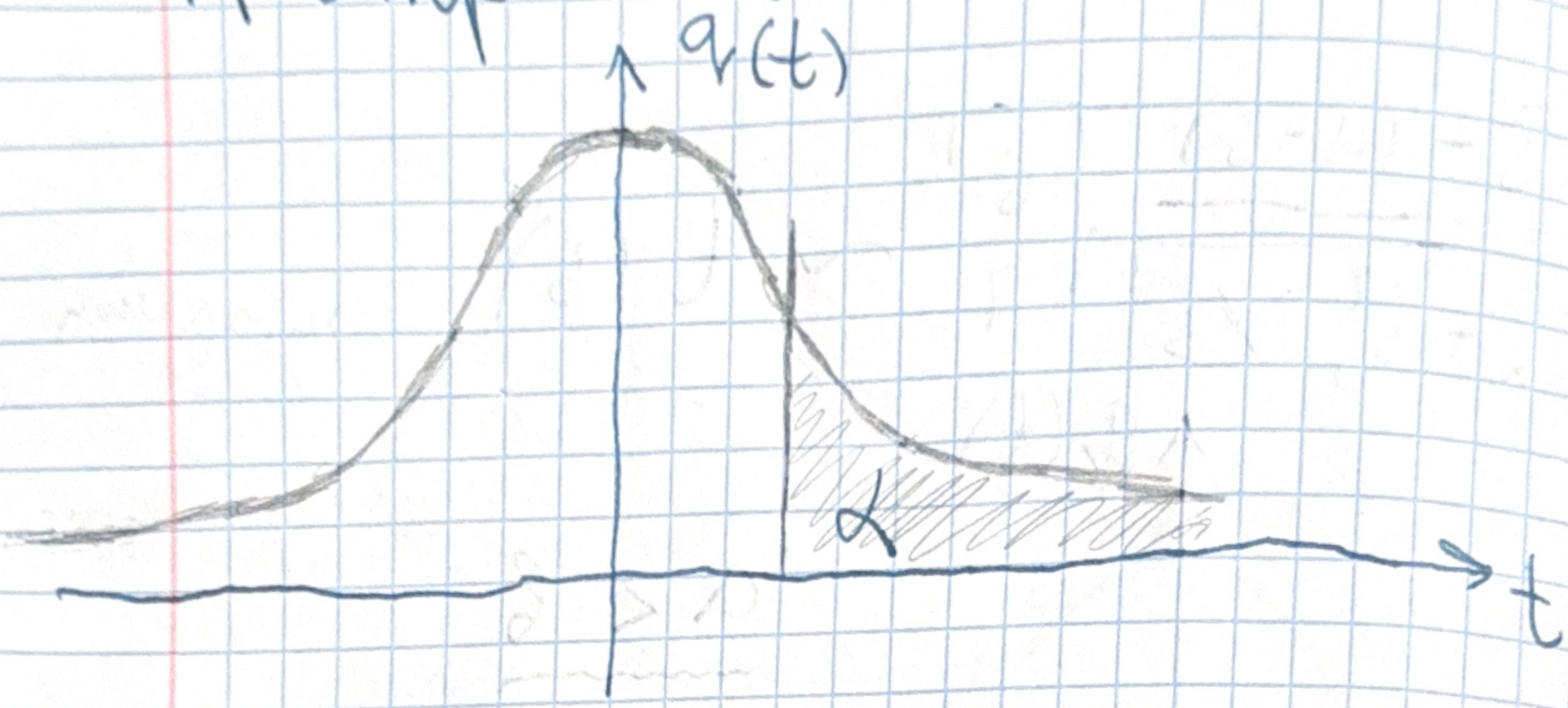
Т Фишера (2 часть)

$$\frac{2 \cdot (n-1) S_x^2}{2} \sim \chi^2(n-1)$$

$$\frac{\frac{1 \cdot S_y^2}{1}}{} \qquad \frac{(n-1) S_x^2}{\sigma_x^2} \sim \chi^2(n-1)$$

$$\frac{(n-1) S_y^2}{\sigma_y^2} \sim \chi^2(n-1)$$

Альтернатива $a > b$



$$G: P(\Delta \geq c \mid H_0) = \alpha$$

$$c = N_{1-\alpha} = 1.645$$

При H_0 $a = b \Rightarrow$

$$\bar{\Delta} = (\bar{x} - \bar{y})\sqrt{\frac{6}{7}} = 0.929 < c$$

$\Rightarrow$ нет веских оснований отвергнуть H_0.

Мощность:

$$W = P(\Delta \geq c \mid H_1) \quad \underline{\underline{a - b = \delta}}$$

$$= P\left(\frac{\bar{x} - \bar{y} - \delta}{\sqrt{\frac{7}{6}}} \geq c - \delta\sqrt{\frac{6}{7}} \right) =$$

$$= P\left(\Delta \geq c - \delta\sqrt{\frac{6}{7}} \right) = \int_{c-\delta\sqrt{\frac{6}{7}}}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \, dx$$